

Roberto (André) Mossi Milla

Tegucigalpa, Honduras • 814-790-1591 • mossiroberto0392@gmail.com • andremossi.vercel.app • github.com/AndreM222

SUMMARY

Software Engineer experienced in AI-powered full-stack platforms, computer vision systems, backend services, and real-time simulation environments. Background includes production web applications, artificial intelligence research, cloud integrations, data processing, and end-to-end system development across healthcare, insurance, industrial automation, and multimodal AI workflows.

WORK EXPERIENCE

Ingeniería Digital, Tegucigalpa, Honduras

Full-Stack Developer / AI Systems Engineer, May 2026 - Present

AI-Assisted Medical Document Generation Platform

- Built an AI-assisted medical document generation platform from the ground up for post-surgery reports (IDR), reducing report preparation from approximately one week to 1–2 days and reducing final document generation to under one minute.
- Combined OCR extraction, LLM-assisted grammar correction, and automated professional document structuring using Next.js, React, TypeScript, PostgreSQL, AWS S3, Amazon Textract, and OpenAI APIs.
- Architected authenticated document management workflows using Auth0, enabling users to upload, generate, analyze, edit, version, review, and manage AI-assisted reports through a full-stack web platform.
- Designed scalable file-processing pipelines integrating OCR extraction, cloud storage, AWS S3, and PostgreSQL persistence for document retrieval and processing.
- Implemented internationalization workflows and automated translation asset generation using CMake to support multilingual deployments across US and Honduran clients.

AI Vehicle Damage Assessment Platform

- Developed an AI-assisted vehicle damage assessment platform that reduced assessment workflows from more than one week to approximately a minute while automatically generating insurance estimates.
- Integrated AWS Rekognition to localize vehicles within uploaded images and Gemini Flash to detect and analyze vehicle damage within the identified vehicle boundaries.
- Integrated multimodal AI, computer vision, VIN-history retrieval, and insurance estimation workflows to support vehicle valuation and claims analysis.
- Developed data acquisition services using Playwright to collect, normalize, and process vehicle information and historical data from external sources.
- Implemented interactive damage annotation and review tools using Konva, allowing users and insurance personnel to identify, visualize, move, and validate detected damage regions.
- Developed an experimental AI evaluation workflow using persisted visual annotations, enabling evaluation of different models, processing stages, and combinations of approaches.
- Developed responsive client-facing and administrative dashboards using React, Next.js, TypeScript, PostgreSQL, AWS S3, and Shadcn/UI.
- Maintained engineering standards and CI/CD workflows using GitHub Actions, Husky, ESLint, Prettier, lint-staged, and Commitizen.

Gannon University, Pennsylvania, United States

Computer Vision Engineer (Project-Based) Industry Partner (NDA), August 2024 - May 2025

- Architected and deployed components of a real-time industrial traceability system using LiDAR-triggered image capture and Ultralytics YOLOv8 to detect and track packages and pallets, automatically generating structured events with images and detection metadata.[4]
- Collaborated in a cross-functional team across hardware, backend, and software components; took primary ownership of the monitoring dashboard and visualization interface.
- Developed a full-stack web dashboard using React and JavaScript, integrated with backend services and TimescaleDB to visualize detection events, images, and operational information in near real time.

- Implemented administrative tools to delete images, move and correct bounding boxes, and curate labeled datasets used for iterative object-detection model retraining.
- Developed image-quality and preprocessing workflows using contrast, color, filtering, spatial image partitioning, and reconstruction techniques to improve image usability within the computer-vision pipeline.
- Engineered an event-driven pipeline synchronizing frame capture with detection metadata, enabling downstream anomaly and damage-analysis workflows.
- Repeatedly retrained and evaluated YOLOv8 using curated datasets, with dataset development reaching approximately 500 images within the project's time constraints.
- Delivered recurring technical presentations and system demonstrations to industry stakeholders and internal reviewers, contributing to architectural decisions including migration from Raspberry Pi hardware to NVIDIA Jetson for improved performance and thermal stability.
- Containerized distributed services using Docker and integrated multi-language components (C++, Python, Rust) to coordinate inference, data processing, and system communication.

Research Assistant, August 2024 - May 2025

- Conducted research under faculty supervision on multi-agent artificial intelligence simulating natural selection, survival behaviors, and emergent behavior in dynamic three-dimensional environments.[8]
- Independently implemented NeuroEvolution of Augmenting Topologies (NEAT) in C++, including the genome representation used to encode neural-network structures and behavior.
- Developed a 3D multi-agent simulation environment in Unreal Engine featuring procedural world generation, physics-based interactions, and evolutionary experimentation using C++, and CUDA
- Developed procedurally generated environments containing halls, floors, walls, and props, enabling experiments across randomized scenarios without manually constructing each environment.
- Research paper accepted and published by the American Society for Engineering Education (ASEE) and presented at a Sigma Xi conference at Penn State University.[5] [3]

Dinant, Tegucigalpa, Honduras

Full-Stack Developer, June - July 2023

- Led the end-to-end design and development of two production web platforms, conducting requirements gathering, system planning, database architecture design, client presentations, and final deployment.[2]
- Architected and implemented a contract lifecycle management system for a device sales, rental, and repair company. Designed relational SQL schemas to track device state (sold, rented, under repair, returned), service eligibility, country-based filtering, and client records across multiple regions.
- Developed a multi-role web application using Oracle APEX, JavaScript, HTML, and CSS, enabling guest contract signing, digital signature capture, automated PDF generation, and administrative dashboards with advanced filtering by date, country, device state, and contract status.
- Engineered a repair workflow system with state transitions, damage logging, cost tracking, time tracking, and daily activity reports to ensure accountability and operational transparency.
- Built a factory maintenance management platform enabling real-time machine status reporting, staged maintenance workflows, automated notifications, repair logs, and material usage tracking with automatic inventory deduction and procurement insight reporting.
- Collaborated directly with international stakeholders, conducted iterative demos, incorporated production feedback, and deployed finalized systems used by operational teams.

INDEPENDENT SOFTWARE & OPEN SOURCE

Copilot-Lualine, Open Source

Creator & Maintainer, 2024 - Present

- Created and maintain an open-source Neovim plugin integrating GitHub Copilot status and controls into the lualine status bar.
- Implemented interactive status indicators and controls for monitoring Copilot state, including loading, authentication, and operational status directly within the editor.
- Maintain the project with community contributions; the project has received 117+ GitHub stars.

EDUCATION

Gannon University, Pennsylvania, United States

Bachelor of Science in Computer Systems Engineering, May 2025[9]

ISI Japanese Language School, Tokyo, Japan

Intensive Japanese Language Program, October 2025 - March 2026

Harvard University, Online

CS50's Introduction to Computer Science, December 2022[10]

Extreme Networks, Online

Introduction to Future Networks, May 2022[1]

PUBLICATIONS

- Mossi, R. A. — “Tag AI-Sandbox,” Proceedings of the ASEE Annual Conference, 2025 (peer-reviewed).[5]

TECHNICAL SKILLS

- Languages: C++, Rust, TypeScript, JavaScript, Python, Java, SQL, Bash, PowerShell, Lua.
- Frontend: React, Next.js, Vite, Tailwind CSS, Shadcn/UI, SwiftUI, Konva.
- Backend & Cloud: Rust, Axum, Node.js, Supabase, PostgreSQL, Redis, TimescaleDB, Oracle Database, AWS S3, Amazon Textract, Docker, Auth0, REST APIs.
- AI & Computer Vision: OpenAI APIs, Gemini, Claude APIs, YOLOv8, CUDA, PyTorch, NEAT, evolutionary computation, OCR pipelines, multimodal AI workflows.
- Developer Tooling: GitHub Actions, Husky, ESLint, Prettier, lint-staged, Commitizen, CMake, Git.
- Operating Systems: Linux, Windows, macOS.
- Languages: Spanish (Native), English (Fluent), Japanese (Intermediate).

AWARDS AND HONORS

- “School Development Award” award obtained for outstanding development of a videogame using Blueprints in Unreal Engine 4, 2020[7]
- “School Development Award” award obtained for outstanding development of a videogame using C++ and Unreal Engine 4, 2016[6]

REFERENCES

- [1] Extreme Academy. *Extreme Networks Certification*. Dec. 1, 2022. URL: <https://andremossi.vercel.app/experiences/extreme> (visited on 02/15/2026).
- [2] Rigoberto Funes. *Dinant Recommendation Letter*. Feb. 11, 2026. URL: <https://andremossi.vercel.app/PDF/Recommendations/dinant.pdf> (visited on 02/15/2026).
- [3] lah585. *Tag AI-Sandbox Presentation*. Sigma Xi. Mar. 11, 2025. URL: <https://sites.psu.edu/behrendsigmaxi2025/2025/03/11/tag-ai-sandbox/> (visited on 02/15/2026).
- [4] Andre Mossi. *Computer Vision Project*. Feb. 15, 2026. URL: <https://andremossi.vercel.app/experiences/traceability> (visited on 02/15/2026).
- [5] Roberto Andre Mossi and Ramakrishnan Sundaram. “Tag AI-Sandbox Paper”. In: *2025 ASEE North Central Section (NCS) Annual Conference*. 10.18260/1-2-54692. Marshall University, Huntington, West Virginia: ASEE Conferences, Mar. 2025. URL: <https://peer.asee.org/54692>.
- [6] Dowal School. *Software Engineer 2016 Award*. May 12, 2016. URL: <https://andremossi.vercel.app/experiences/dowal2016> (visited on 02/15/2026).
- [7] Dowal School. *Software Engineer 2020 Award*. May 12, 2020. URL: <https://andremossi.vercel.app/experiences/dowal2020> (visited on 02/15/2026).
- [8] Ram Sundaram. *Tag AI-Sandbox Recommendation Letter*. Feb. 20, 2025. URL: <https://andremossi.vercel.app/PDF/Recommendations/ai-sandbox.pdf> (visited on 02/15/2026).
- [9] Gannon University. *Gannon University BS Diploma*. May 10, 2025. URL: <https://andremossi.vercel.app/experiences/gannon> (visited on 02/15/2026).

- [10] Harvard University. *Harvard CS50 Certification*. Dec. 20, 2022. URL: <https://andremossi.vercel.app/experiences/harvard> (visited on 02/15/2026).